

# Rewiring the market time clock in Spanish electricity markets: strategic behaviour and market power with higher granularity

Pablo Paramio Pérez  
CEMFI Master in Economics and Finance

June 2026

**Abstract**

## Glossary of acronyms

| Acronym | Meaning                                                                                                                    |
|---------|----------------------------------------------------------------------------------------------------------------------------|
| CCGT    | Combined-cycle gas turbine: a flexible, dispatchable thermal generator.                                                    |
| DA      | Day-ahead wholesale electricity market, cleared once per day.                                                              |
| IDA     | Intraday auctions: sequence of (European) auctions held after the day-ahead clearing.                                      |
| MTU     | Market time unit: size of the time interval at which bids are submitted and prices clear.                                  |
| REE     | Red Eléctrica de España, the Spanish transmission-system operator.                                                         |
| OMIE    | Operador del Mercado Ibérico de Energía, the market operator for the Spanish and Portuguese wholesale electricity markets. |

# 1 Introduction

## 1.1 Motivation and research question

## 1.2 Related literature

## 1.3 Mechanism summary and contribution

TWO DISTINCT PURPOSES OF MTU15:

- OPERATIONAL / SYSTEM-OPERATOR CHANNEL. LETTING THE SYSTEM FOLLOW WITHIN-HOUR SOLAR VARIABILITY (CLOUD COVER, RAMP). MIDDAY IS EXACTLY WHERE THIS MATTERS MOST.
- STRATEGIC-EXPLOITATION CHANNEL. LETTING PIVOTAL FIRMS RE-PRICE ACROSS QUARTERS WHEN NET-LOAD MOVES SYSTEMATICALLY WITHIN THE HOUR. THIS IS WHAT DOMINANT CCGT CAN MONETIZE. **THIS IS WHAT WE DO!!**

## 1.4 Roadmap

# 2 Theoretical model

## 2.1 Setup

## 2.2 Equilibrium under low granularity

## 2.3 Equilibrium under high granularity

## 2.4 Predictions: dominant vs fringe response in time-varying vs flat subperiods

TODO: COROLLARY ON RAMP-CONSTRAINED HOURS. WHEN THE SYSTEM OPERATOR IMPOSES BINDING RAMP RESTRICTIONS (QUANTITY CANNOT ADJUST MUCH ACROSS CONSECUTIVE QUARTERS), THE HIGH-GRANULARITY REGIME FORCES FIRMS TO AMPLIFY THEIR STRATEGIC RESPONSE ON THE PRICE MARGIN RATHER THAN THE QUANTITY MARGIN. THE MTU15 REFORM THEREFORE AMPLIFIES MARKET POWER IN PRECISELY THOSE HOURS WHERE PHYSICAL ADJUSTMENT IS CONSTRAINED.

# 3 Institutional setting, data, and identification

## 3.1 Spanish wholesale electricity market: a sequential-market overview

STYLISTED MARKET DIAGRAM TBD HERE AS TIKZ: DA - IDA AUCTIONS - CONTINUOUS - BALANCING

## 3.2 The MTU15 reform sequence

## 3.3 Data sources

OMIE

- Day-ahead data:

- Intraday-auction data:
- Continuous intraday market data:

ESIOS

ENTSO-E

INCLUDE SAMPLE SIZE AND FAMILIES.

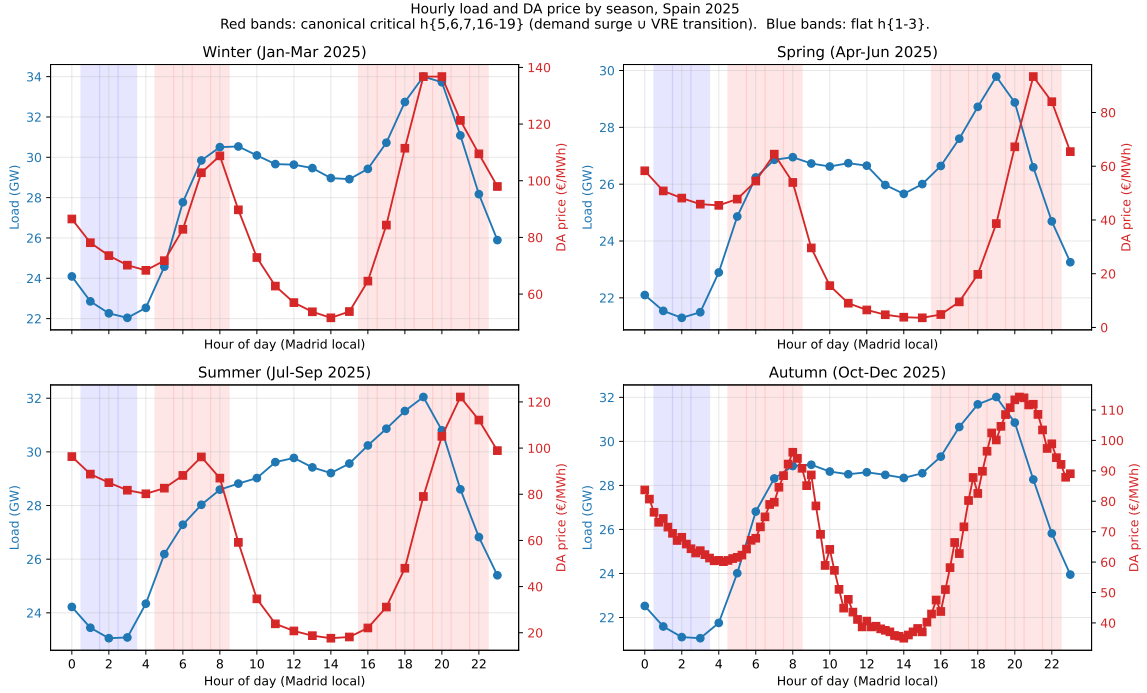
### 3.4 Critical-vs-flat hours: empirical calibration

Hours are classified by the **within-hour variance of residual demand**,

$$\sigma_{\text{within}}^2(h) = \mathbb{E}_d \left[ \frac{1}{4} \sum_{q=1}^4 (D_{h,q,d} - \bar{D}_{h,d})^2 \right], \quad D_{h,q,d} = \text{demand}_{h,q,d} - \text{wind}_{h,q,d} - \text{solar}_{h,q,d}. \quad (1)$$

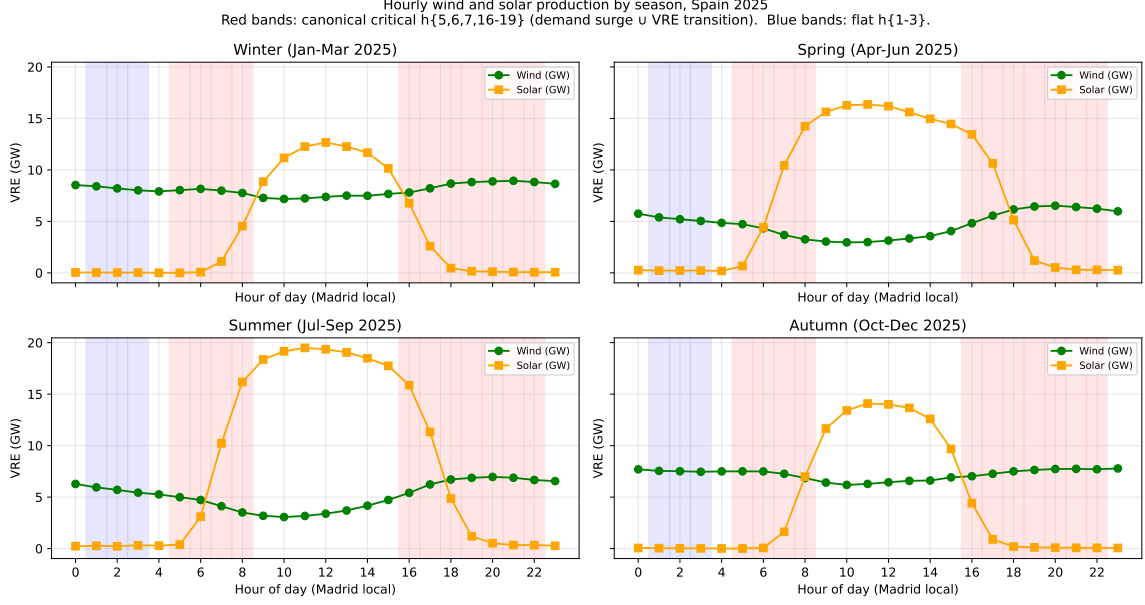
**Critical hours:** high  $\sigma_{\text{within}}^2$ , 05:00–09:00 and 16:00–23:00 (eleven clock-hours). **Flat hours:** low  $\sigma_{\text{within}}^2$ , 01:00–04:00 (three clock-hours). Borderline hours are dropped. Full 24-hour ranking in Appendix B.6.

OJO, IF WE INTRODUCE THE 11:00-15:00 PERIOD IN THE CONTROL, WE ARE MIXING THE OPERATIONAL AND STRATEGIC CHANNELS OF THE REFORM, BECAUSE THE OPERATIONAL EFFECT IS CONCENTRATED IN THOSE HOURS!!!



**Figure 1: Hourly load and DA price by season, Spain 2025.** Blue: demand; red: DA clearing price. Red bands: critical hours (05:00–09:00, 16:00–23:00); blue bands: flat hours (01:00–04:00).

*Reading (supportive).* Critical hours combine the morning ramp and evening peak; flat hours fall in the overnight valley. The pattern is stable across seasons.



**Figure 2: Hourly wind and solar production by season, Spain 2025.** Red bands: critical hours; blue bands: flat hours.

*Reading (supportive).* Solar production concentrates around midday and fades sharply at the critical-hour boundaries; wind is broadly flat. Critical hours systematically face low renewable supply, reinforcing their role as the time-varying window.

### 3.5 Firms in the analysis sample

MENTION THAT WE USE LISTA\_UNIDADES AND LISTA\_AGENTES FROM OMIE, AS WELL AS THE SPECIFIC FAMILIES FORM OMIE, WHICH HAS FIRM AND UNIT INDICATORS. ALSO IMPORTANT TO MENTION THE CNMC CLASSIFICATION OF DOMINANT (SOURCE)

The empirical analysis follows nine parent firms in two blocks. The first block contains the *dominant operators in Spanish generation* (the CNMC *operador-dominante-en-generación* tier): Iberdrola, Endesa, Naturgy and EDP-Spain. EDP-Portugal joins this block as the dominant Iberian generator on the Portuguese side of MIBEL. The second block contains four other large vertically-integrated operators – Repsol, Engie España, TotalEnergies, and Moeve – which retain substantial portfolios but are non-dominant in the CNMC sense. Whether the firms in each block actually have a strategic margin in critical hours is an empirical question that we answer in Section 4.1, examining pivotality on the day-ahead bid data.

**Table 1: Firms in the analysis sample.** Capacity: max DA-offered power across 2025, share-weighted. Market share: PDBF programmed sell, 2025. Channel breakdown in Appendix B.2.

| Parent firm                                                 | Main generation technologies | Capacity (GW) | Market share (%) |
|-------------------------------------------------------------|------------------------------|---------------|------------------|
| <i>Dominant operators in Spanish generation (CNMC tier)</i> |                              |               |                  |
| Iberdrola                                                   | Solar, Hydro, CCGT           | 31.03         | 24.94            |
| Endesa                                                      | Hydro, CCGT, Nuclear         | 12.46         | 16.39            |
| Naturgy                                                     | CCGT, Hydro, Solar           | 12.60         | 12.34            |
| EDP-Spain                                                   | Wind, Hydro, Solar           | 4.73          | 4.14             |
| EDP-Portugal                                                | Hydro, CCGT                  | 7.42          | 15.17            |
| <i>Other large operators (non-dominant)</i>                 |                              |               |                  |
| Repsol                                                      | Solar, Hydro                 | 0.61          | 1.43             |
| Engie España                                                | Wind, Solar, CCGT            | 4.05          | 1.73             |
| TotalEnergies                                               | Solar, CCGT, Wind            | 1.19          | 0.63             |
| Moeve                                                       | Solar, Hydro, CCGT           | 1.28          | 0.01             |

*Reading (descriptive).* The four CNMC-dominant operators (Iberdrola, Endesa, Naturgy, EDP-Spain) plus EDP-Portugal together hold the bulk of Iberian generation capacity. The non-dominant block (Repsol, Engie, TotalEnergies, Moeve) is substantially smaller and serves as the placebo.

### 3.6 Identification strategy and treatment-effect set

WE SHOULD DEFINE HERE HOW WE DEFINE PIVOTALITY AND HOW WE CALCULATE IT, WITH DISCUSSION.

ALSO WE SHOULD DISCUSS THAT THE TREATMENT IN OUR IDENTIFICATION IS AT THE HOUR LEVEL (CRITICAL VS FLAT), NOT AT THE FIRM LEVEL, AND THAT FIRMS ENTER THE ANALYSIS AS SUBJECTS: PIVOTAL FIRMS ARE THOSE EXPECTED TO REACT TO THE TREATMENT BECAUSE THEY HAVE A STRATEGIC MARGIN IN CRITICAL HOURS; NON-PIVOTAL FIRMS SERVE AS PLACEBO BECAUSE THEY BID AT MARGINAL COST REGARDLESS OF HOUR.

HOWEVER, WE SHOULD THINK IF THIS IS THE BEST WAY TO PRESENT IT, MAYBE IT IS BETTER TO TALK ABOUT "EXPOSURE" TO THE TREATMENT ACROSS TIME OR SOMETHING LIKE THAT!! TO THINK ABOUT IT

### 3.7 Parallel-trends discussion

IMPORTANT THING HERE IS TO DISCUSS IT FOR EACH OF THE OUTCOMES AND TO TALK ABOUT CONDITIONAL PARALLEL TREND WITH RENEWABLE CAPACITY CONTROLS!

## 4 Bidding-behavior evidence of the mechanism

### 4.1 Pivotality: which firms can exploit the granularity reform

OJO, HYDRO DOES NOT EXPLOIT GRANULARITY BECAUSE THE DISPATCH DECISION IS DOMINATED BY RESERVOIR WATER-VALUE OPTIMIZATION (A MULTI-DAY PROBLEM), NOT WITHIN-HOUR STRATEGY. SO THE GRANULARITY REFORM'S "STRATE-

GIC EXPLOITATION CHANNEL” HAS NO PURCHASE HERE EVEN THOUGH THE CAPACITY FOR MARKET POWER EXISTS!!!!

ONLY CCGT EXPLOITS GRANULARITY!! AND IN PARTICULAR NATURGY!

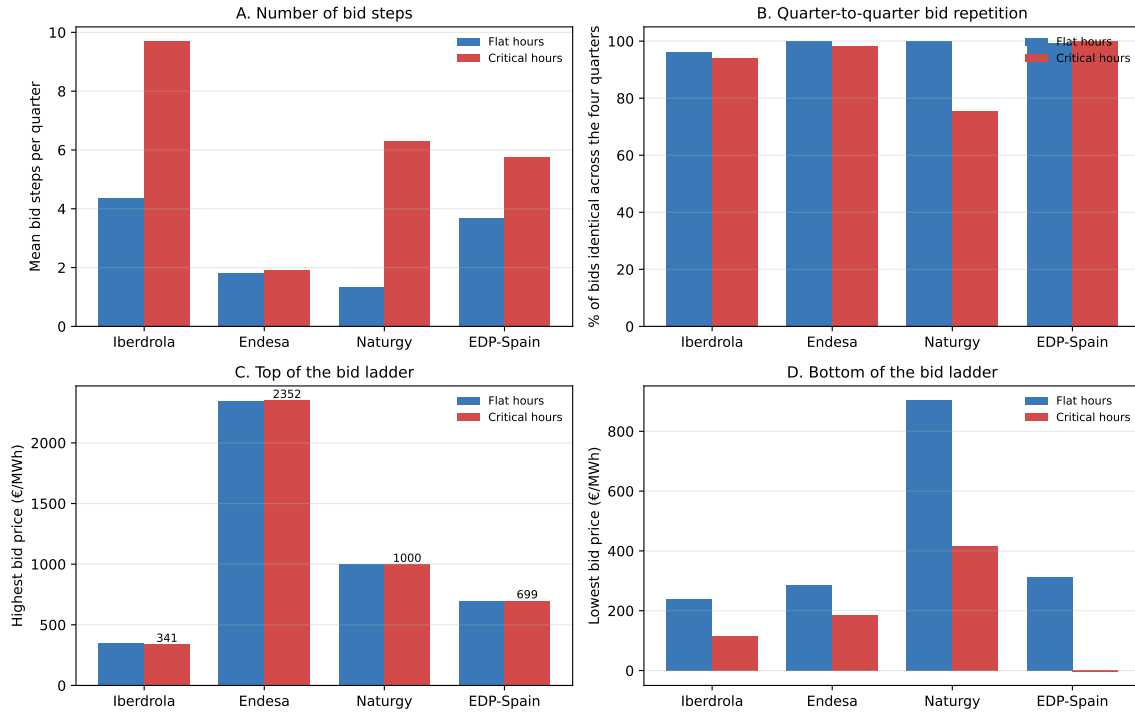
**Table 2: Pivotality by parent firm and hour-class, CCGT, October–December 2025.** Pivotality = share of (date, unit, period) cells with a bid tranche within 1 EUR/MWh of DA clearing.

| Firm (parent)                                                           | flat (01:00–04:00) | critical (05:00–09:00, 16:00–23:00) | $\Delta$ crit–flat |
|-------------------------------------------------------------------------|--------------------|-------------------------------------|--------------------|
| <i>Tightly pivotal in critical hours (within 1 EUR/MWh of clearing)</i> |                    |                                     |                    |
| Naturgy                                                                 | 0.006              | 0.044                               | <b>0.038</b>       |
| EDP-Spain                                                               | 0.026              | 0.041                               | <b>0.015</b>       |
| EDP-Portugal                                                            | 0.005              | 0.028                               | 0.023              |
| Iberdrola                                                               | 0.003              | 0.012                               | 0.009              |
| Endesa                                                                  | 0.001              | 0.004                               | 0.003              |
| <i>Essentially never tightly pivotal (negative control)</i>             |                    |                                     |                    |
| Engie España                                                            | 0.000              | 0.000                               | 0.000              |
| Moeve (formerly Cepsa)                                                  | 0.000              | 0.000                               | 0.000              |
| Repsol                                                                  | 0.000              | 0.000                               | 0.000              |
| TotalEnergies                                                           | 0.000              | 0.000                               | 0.000              |

*Reading (supportive).* Naturgy is the most pivotal firm in critical hours (4.4%), followed by EDP-Spain and EDP-Portugal. Repsol, Engie, TotalEnergies, and Moeve are essentially never tightly pivotal and serve as the negative control.

## 4.2 Per-firm bid-shape and granularity exploitation

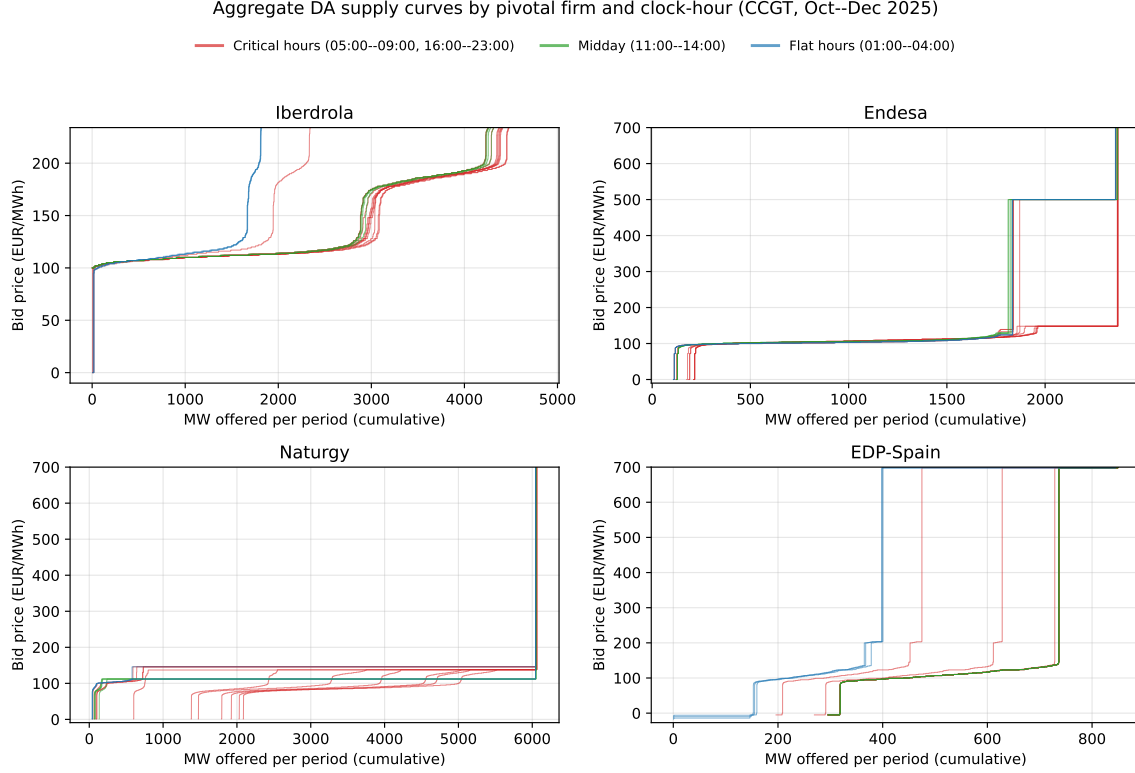
Per-firm bid behaviour, critical vs flat hours



A bid step is one price-quantity pair in a firm's supply offer for a given period; a firm with more steps offers a finer-grained supply curve. 'Identical across quarters' = the firm submitted the exact same step (price and quantity) in all four quarters of the same clock-hour. Sample: combined-cycle gas (CCGT) units operated by the four largest Iberian groups, day-ahead market, October–December 2025 (post-reform window).

**Figure 3: Per-firm DA bid behaviour, critical vs flat hours.** Top-4 CCGT operators, October–December 2025. Panel A: number of price-quantity steps per 15-minute period. Panel B: share of bids identical across all four quarters of the same hour. Panels C–D: highest and lowest priced step.

*Reading (supportive).* Two distinct strategic patterns appear. Iberdrola enriches the bid ladder in critical hours (more steps) but keeps the four quarters of each hour nearly identical. Naturgy keeps the ladder shallow and varies bids strongly across quarters.



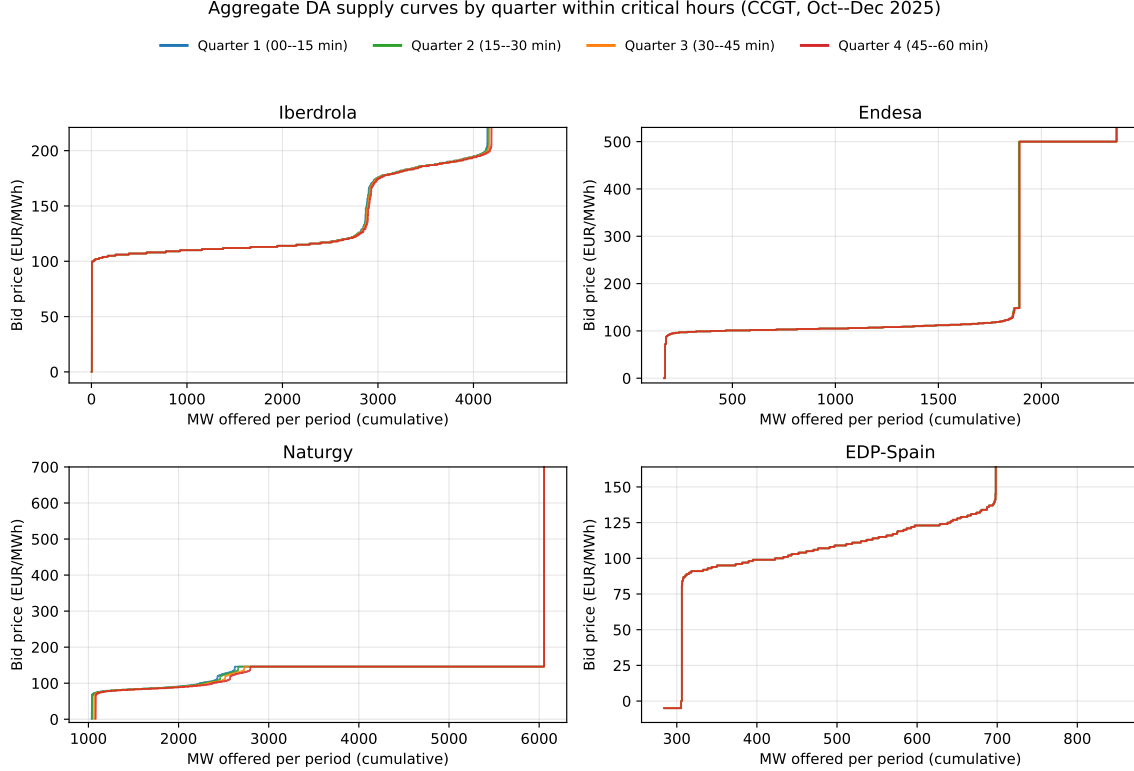
**Figure 4: Aggregate DA supply curves by pivotal firm and clock-hour, CCGT, October–December 2025.** EUPHEMIA aggregation (public description, sec. 7.1) per (firm, hour): pool sell tranches, sort ascending by price, cumulate, normalise by  $N_{\text{dates}} \times 4$  quarters. Red: 11 critical hours; green: 4 midday hours (11:00–14:00, dropped from the DiD); blue: 3 flat hours. Submission rates by hour-class in Table 9; other technologies in Appendix A.10.

*Reading (supportive).* Within-day variation is what we look at: the spread of red curves within a panel shows whether the firm differentiates its bid hour-by-hour in critical hours. Iberdrola’s red curves spread across a wide range while the three blue flat-hour curves sit tightly together; midday (green) sits between the two. Naturgy’s pattern is sharper: critical-hour curves span  $\sim 1,500$ – $6,000$  MW at moderate prices, flat curves cluster at the lower edge. EDP-Spain shows the same within-class fan in critical hours. Endesa is the exception: red, green, and blue curves overlap and do not spread — no within-day differentiation.

**Table 3: Per-firm DA bid-shape detail, October–December 2025.** Mean number of tranches, median tranche price, total quantity offered, and the share of clock-hours in which all four 15-min quarters carry an identical bid (mechanical-repeat rate).

| Firm         | Hour-class | Mean tranches | Median price (EUR/MWh) | Total qty (MW) | Mechanical repeat rate | N      |
|--------------|------------|---------------|------------------------|----------------|------------------------|--------|
| Iberdrola    | Critical   | 9.70          | 134.8                  | 612.4          | 0.990                  | 31,324 |
|              | Flat       | 4.38          | 245.7                  | 614.4          | 0.969                  | 8,556  |
| Endesa       | Critical   | 1.92          | 1090.0                 | 460.7          | 0.982                  | 20,780 |
|              | Flat       | 1.82          | 1179.4                 | 460.6          | 1.000                  | 5,652  |
| Naturgy      | Critical   | 6.30          | 538.1                  | 410.4          | 0.811                  | 59,720 |
|              | Flat       | 1.32          | 945.5                  | 410.4          | 1.000                  | 16,260 |
| EDP-Spain    | Critical   | 5.76          | 121.1                  | 426.8          | 1.000                  | 8,052  |
|              | Flat       | 3.68          | 375.9                  | 424.9          | 1.000                  | 2,196  |
| EDP-Portugal | Critical   | 6.60          | 153.0                  | 407.0          | 0.075                  | 18,556 |
|              | Flat       | 5.52          | 233.0                  | 407.0          | 0.270                  | 5,064  |

*Reading (supportive).* Ladder enrichment in critical hours is clear in Iberdrola (9.7 vs 4.4 tranches), Naturgy (6.3 vs 1.3) and EDP-Spain (5.8 vs 3.7). The mechanical-repeat rate drops in critical hours for Naturgy (0.81 vs 1.00) and EDP-Portugal (0.07 vs 0.27): they are the firms varying their bids across the four quarters of an hour. The higher flat-hour median prices are mechanical — low overnight demand pushes units into precautionary, near-cap bids regardless of strategy.



**Figure 5: Aggregate DA supply curves by quarter within critical hours, CCGT.** EUPHEMIA aggregation per (firm, quarter) over the 11 critical hours and all days in October–December 2025. One curve per quarter.

*Reading (mixed).* On average over critical hours the four quarter-curves nearly coincide for all four firms — consistent with the mechanical-repeat rates in Table 3. Naturgy shows the largest within-hour quarter dispersion, visible around 2,400–2,900 MW. Granularity is exploited episodically, not as a systematic four-quarter bid plan. Hydro, Wind and Solar PV per-quarter analogues are in Appendix A.10.

## 5 Main empirical results

### 5.1 Strategic withholding and intraday upward repositioning

MECHANISM: UNDER THE HIGHER-GRANULARITY REGIME, A PIVOTAL DOMINANT FIRM IN A CRITICAL HOUR CAN WITHHOLD CAPACITY FROM THE DAY-AHEAD CLEARING AND REPOSITION THAT CAPACITY UPWARD IN THE INTRADAY AUCTIONS AT HIGHER PER-QUARTER PRICES. UPWARD SELL ADJUSTMENT SUMMED ACROSS THE THREE INTRADAY AUCTIONS,  $q_{2,fdh}$ , IN MWH PER UNIT-CLOCK-HOUR:

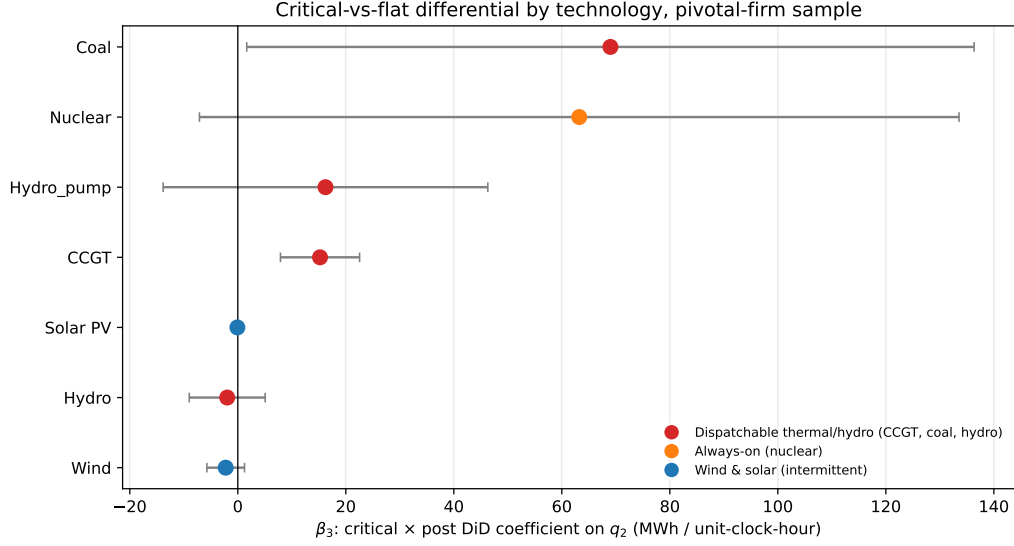
$$q_{2,fdh} = \alpha + \beta_1 \text{crit}_h + \beta_2 \text{post}_d + \beta_3 (\text{crit}_h \times \text{post}_d) + \gamma_f + \delta_{\text{DOW}(d)} + \varepsilon_{fdh}, \quad (2)$$

with firm fixed effects  $\gamma_f$ , day-of-week fixed effects  $\delta_{\text{DOW}(d)}$ , and cluster-robust standard errors by date. The sample restricts to the same-calendar-month windows October–December 2024 (pre) and October–December 2025 (post), so the March 2025 intraday reform is held constant. Critical hours are 05:00–09:00 and 16:00–23:00; flat hours are 01:00–04:00. Pivotal firms are those with non-trivial price-setting capacity in critical hours (Iberdrola, Endesa, Naturgy, EDP-Spain, EDP-Portugal); non-pivotal firms (Repsol, Engie España, TotalEnergies, Moeve) serve as a negative control.

**Table 4: B1 – Within-day DiD on  $q_2$  (intraday upward adjustment).** Estimates of Equation 2. Significance: \*  $p < 0.05$ , \*\*  $p < 0.01$ , \*\*\*  $p < 0.001$ .

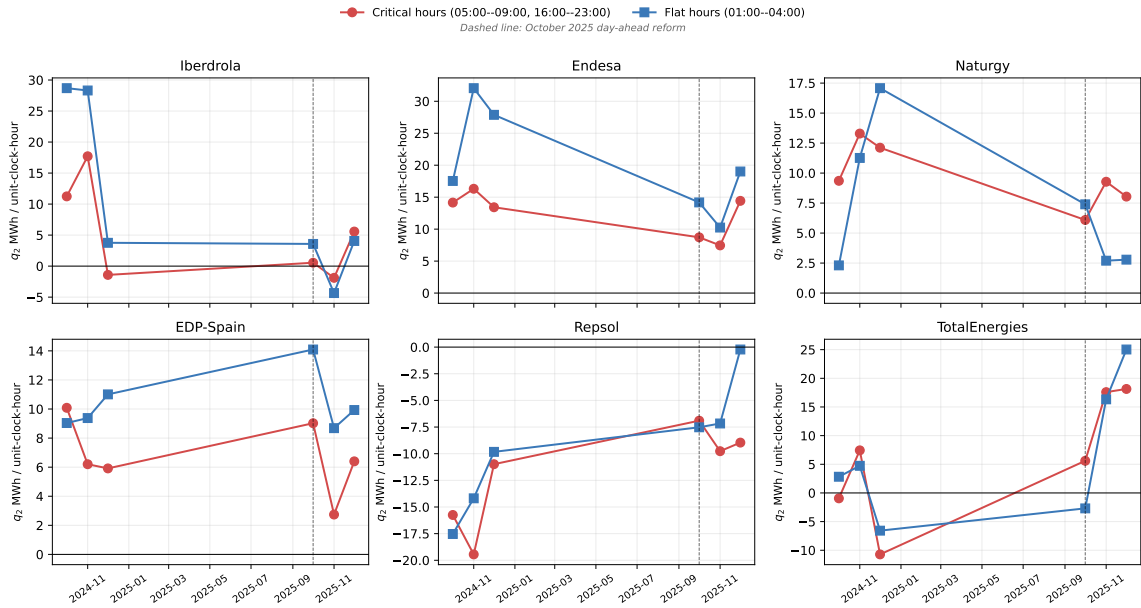
|                                      | All firms<br>(1)     | Pivotal firms<br>(2) | Non-pivotal firms<br>(3) |
|--------------------------------------|----------------------|----------------------|--------------------------|
| Critical ( $\beta_1$ )               | 1.280***<br>(0.328)  | -4.454***<br>(0.893) | 4.993***<br>(0.320)      |
| Post ( $\beta_2$ )                   | -2.562***<br>(0.486) | -8.294***<br>(1.146) | 1.140**<br>(0.415)       |
| Critical $\times$ Post ( $\beta_3$ ) | -1.397**<br>(0.446)  | 3.866***<br>(1.147)  | -4.741***<br>(0.423)     |
| Observations                         | 293,838              | 122,495              | 171,343                  |
| Clusters (days)                      | 184                  | 184                  | 184                      |
| $R^2$                                | 0.0132               | 0.0056               | 0.0113                   |
| Adj. $R^2$                           | 0.0131               | 0.0055               | 0.0113                   |

*Reading (supportive).*  $\beta_3 = +3.87$  MWh per unit-clock-hour for pivotal firms and  $-4.74$  for the non-pivotal placebo, both significant at the 0.1% level. The sign reverses across firm classes, which the placebo prediction requires.



**Figure 6: B1 –  $\beta_3$  by technology, pivotal sample.** Equation 2 restricted by `tech_group`; 95% cluster-robust CIs.

*Reading (mixed).* Positive  $\beta_3$  concentrates in the strategically dispatchable techs (CCGT +15.2, Coal +69, Hydro\_pump +16.2); Wind and Solar PV both show the predicted null. Hydro is mildly negative against a positive prediction, consistent with reservoir water-value optimisation rather than within-hour strategy. Nuclear is large with very wide error bars and few units — neither sharply contradicts the mechanism nor cleanly supports it.



**Figure 7:  $q_2$  monthly mean by firm and hour-class.** Red: critical (05:00–09:00, 16:00–23:00); blue: flat (01:00–04:00). Dashed line: October 2025 reform.

*Reading (supportive).* The pivotal-firm panels show the critical line tracking above the flat line after October 2025, while the non-pivotal panels show the two lines collapsing together. The pre-reform trends are visually parallel in both blocks.

The full table of  $\beta_3$  stratified by technology (with the non-pivotal placebo column) is reported in Appendix A.4. The per-firm decomposition of  $\beta_3$  jointly on  $q_2$  and on day-ahead cleared volume (the withholding outcome) is in Appendix A.5. A supporting analysis on the day-ahead

→ intraday price wedge is in Appendix A.6; the wedge is a noisy outcome under the post-blackout *operación reforzada* regulatory overlay and is not informative on its own.

## 5.2 Parallel trends discussion

$$\mathbb{E}[Y_{\text{post}}(0) - Y_{\text{pre}}(0) \mid \text{crit}] = \mathbb{E}[Y_{\text{post}}(0) - Y_{\text{pre}}(0) \mid \text{flat}]. \quad (3)$$

# 6 Conclusion

## A Robustness checks

### A.1 Sensitivity to critical-hours definition

### A.2 Sensitivity to firm-partition (pivotality vs administrative)

### A.3 Same-cal-month vs full-window comparisons

**Table 5: B5 –  $\beta_3$  across sensitivities, pivotal firms on  $q_2$ .** Each row re-estimates Equation 2 under one variation: critical-hours definition (B5.1), firm partition (B5.2), window (B5.3/B5.5), sample exclusions (B5.4), DST transition days (B5.6/B5.7).

| Sensitivity                                                                       | $\beta_3$ | SE     | $p$    | $G$ |
|-----------------------------------------------------------------------------------|-----------|--------|--------|-----|
| <i>B5.1 — Critical-hours definition</i>                                           |           |        |        |     |
| supply_ramp h{7,8,16,17,18}                                                       | 4.28***   | (1.26) | 0.0007 | 184 |
| price_peak h{18-22}                                                               | 5.68***   | (1.49) | 0.0001 | 184 |
| demand_peak h{16-20}                                                              | 5.13***   | (1.36) | 0.0002 | 184 |
| joint h{7-8, 16-22}                                                               | 4.77***   | (1.31) | 0.0003 | 184 |
| <i>B5.2 — Firm partition</i>                                                      |           |        |        |     |
| Pivotality-based firm set                                                         | 3.87***   | (1.15) | 0.0008 | 184 |
| Administrative IB/GE/GN/HC                                                        | 3.87***   | (1.15) | 0.0008 | 184 |
| <i>B5.3 / B5.5 — Window</i>                                                       |           |        |        |     |
| Full window 2024-2025                                                             | 3.42***   | (0.80) | 0.0000 | 730 |
| Full window minus Apr-Sep 2025                                                    | 4.00***   | (0.85) | 0.0000 | 575 |
| <i>B5.4 — Sample exclusions</i>                                                   |           |        |        |     |
| Drop EDP-PT                                                                       | 3.87***   | (1.15) | 0.0008 | 184 |
| Drop ABO2G                                                                        | 3.87***   | (1.15) | 0.0008 | 184 |
| <i>B5.6 — DST transition days (CET ↔ CEST)</i>                                    |           |        |        |     |
| Same-cal-month, drop DST transition days                                          | 4.00***   | (1.15) | 0.0005 | 182 |
| Full window, drop DST transition days                                             | 3.47***   | (0.80) | 0.0000 | 726 |
| <i>B5.7 — DST regime separation (clock-hour semantics differ across CEST/CET)</i> |           |        |        |     |
| Same-cal-month, CEST days only (Oct 1–25)                                         | −2.45     | (1.93) | 0.2028 | 50  |
| Same-cal-month, CET days only (Oct 27–Dec 31)                                     | 6.30***   | (1.34) | 0.0000 | 132 |

*Reading (supportive).*  $\beta_3$  stays in the +3.4 to +6.3 MWh range across critical-hours definitions, firm partitions, window choices, and DST handling, with  $p < 0.001$  throughout. The only sub-window where the sign flips is the 50-day CEST-only segment, which is small and confounded by daylight saving.

#### A.4 $\beta_3$ stratified by technology

**Table 6: B1 –  $\beta_3$  by technology, pivotal sample.** Equation 2 restricted by `tech_group`.

|                                      | CCGT                | Hydro             | Hydro pump       | Coal              | Nuclear          | Wind              | Solar PV |
|--------------------------------------|---------------------|-------------------|------------------|-------------------|------------------|-------------------|----------|
| Critical $\times$ Post ( $\beta_3$ ) | 15.23***<br>(3.738) | -1.963<br>(3.586) | 16.24<br>(15.34) | 69.00*<br>(34.36) | 63.22<br>(35.88) | -2.244<br>(1.779) | 0<br>(.) |
| Obs.                                 | 25,232              | 16,997            | 3,125            | 413               | 2,440            | 31,344            | 7,276    |
| Clusters                             | 184                 | 184               | 183              | 58                | 118              | 184               | 184      |

*Reading (mixed).* Same picture as Figure 6 in numeric form: dispatchable carry the result, Wind and Solar PV are null, Nuclear is large but extremely noisy. Hydro pushes against the directional prediction, in line with reservoir water-value optimisation rather than within-hour strategy.

#### A.5 Per-firm decomposition: $q_2$ and day-ahead cleared MWh

**Table 7: Per-firm  $\beta_3$  on  $q_2$  and DA cleared MWh.**

| Firm         | $q_2$ outcome (B1) |        | DA cleared (B3) |         |
|--------------|--------------------|--------|-----------------|---------|
|              | $\beta_3$          | SE     | $\beta_3$       | SE      |
| Iberdrola    | 10.40**            | (3.25) | -62.37***       | (14.02) |
| Endesa       | 7.30*              | (3.06) | -2.22           | (6.57)  |
| Naturgy      | 2.58               | (1.79) | -34.69***       | (7.69)  |
| EDP-Spain    | -2.29              | (1.53) | -0.31           | (3.33)  |
| EDP-Portugal | —                  | —      | -24.18*         | (11.43) |

*Reading (mixed).* Iberdrola and Naturgy show the joint pattern predicted by the mechanism (negative DA, positive IDA). Endesa adds in IDA without DA withholding, and EDP-Spain is weak on both sides — the mechanism is not uniform across pivotal firms. EDP-Portugal lacks `pibci` data so only the DA side is identified for that firm.

#### A.6 DA $\rightarrow$ IDA price wedge (supporting outcome)

**Table 8: B2 – Within-day DiD on the DA-IDA price wedge.** Outcome in EUR/MWh.

| Specification                                                                             | $\beta_3$ | SE     | $p$    | $G$ |
|-------------------------------------------------------------------------------------------|-----------|--------|--------|-----|
| Same-cal-month (Oct-Dec 2024 vs 2025)                                                     | 4.62**    | (1.43) | 0.0012 | 184 |
| Full window, no controls                                                                  | 0.79      | (1.18) | 0.5043 | 730 |
| Full window, + <code>reforzada</code> $\times$ <code>crit</code>                          | -10.06*** | (1.39) | 0.0000 | 730 |
| Full window, + <code>reforzada</code> + <code>MTU15-IDA</code> $\times$ <code>crit</code> | -10.06*** | (1.39) | 0.0000 | 730 |

*Reading (problematic).* Under same-calendar-month restriction the within-day wedge expands modestly post-reform, but once the `reforzada`  $\times$  `critical` interaction is included in the full-window spec the reform’s own coefficient turns large and negative. The wedge is not informative without controlling for the regulatory overlay, so we cannot use it as a clean supporting outcome.

## A.7 Operational vs strategic decomposition (PIBCA vs PHF)

## A.8 Bid-shape robustness in the competitive-zone restriction

## A.9 DA submission rates by hour-class (selection check)

Table 9: Mean per-unit CCGT DA submission rate, by firm and hour-class, October–December 2025.

| Firm         | Critical | Midday | Flat  | N units |
|--------------|----------|--------|-------|---------|
| Iberdrola    | 0.860    | 0.861  | 0.861 | 9       |
| Endesa       | 0.733    | 0.734  | 0.731 | 7       |
| Naturgy      | 0.868    | 0.868  | 0.866 | 17      |
| EDP-Spain    | 0.995    | 0.995  | 0.995 | 2       |
| EDP-Portugal | 0.917    | 0.915  | 0.917 | 5       |

*Reading (supportive).* Within each firm, critical / midday / flat rates differ by  $< 0.003$ . No hour-class selection: same units bid across hour classes.

## A.10 Per-firm aggregate supply curves, other technologies

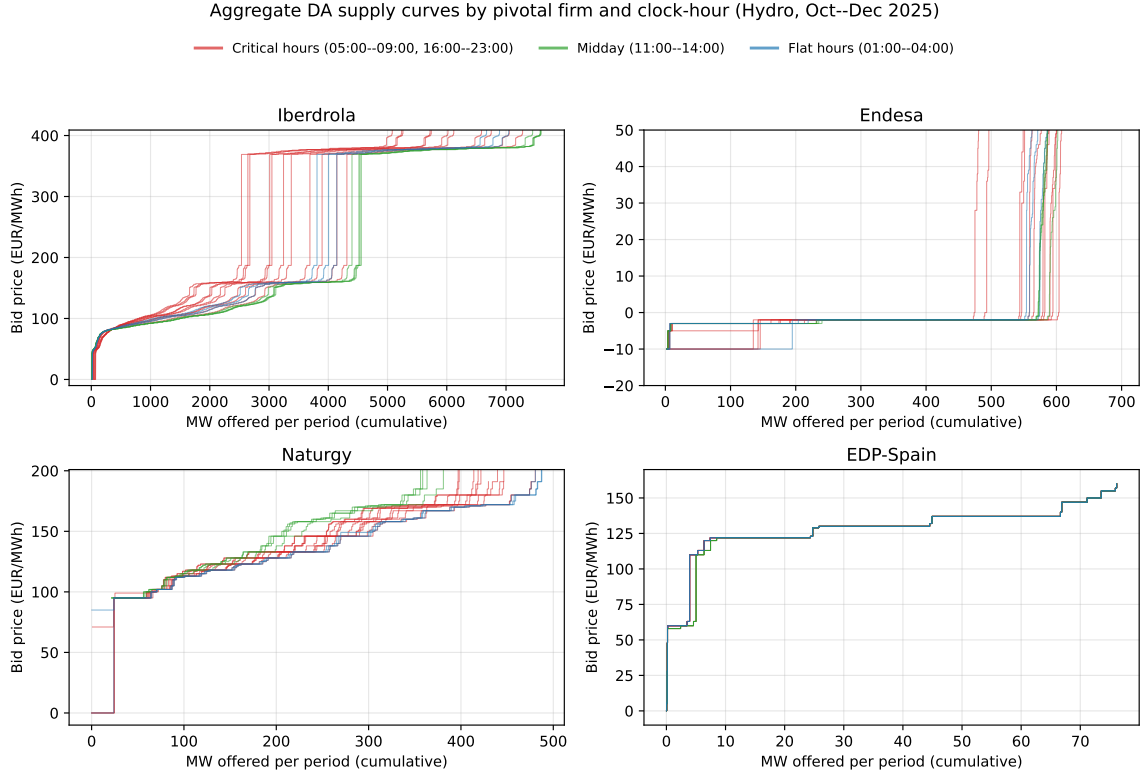
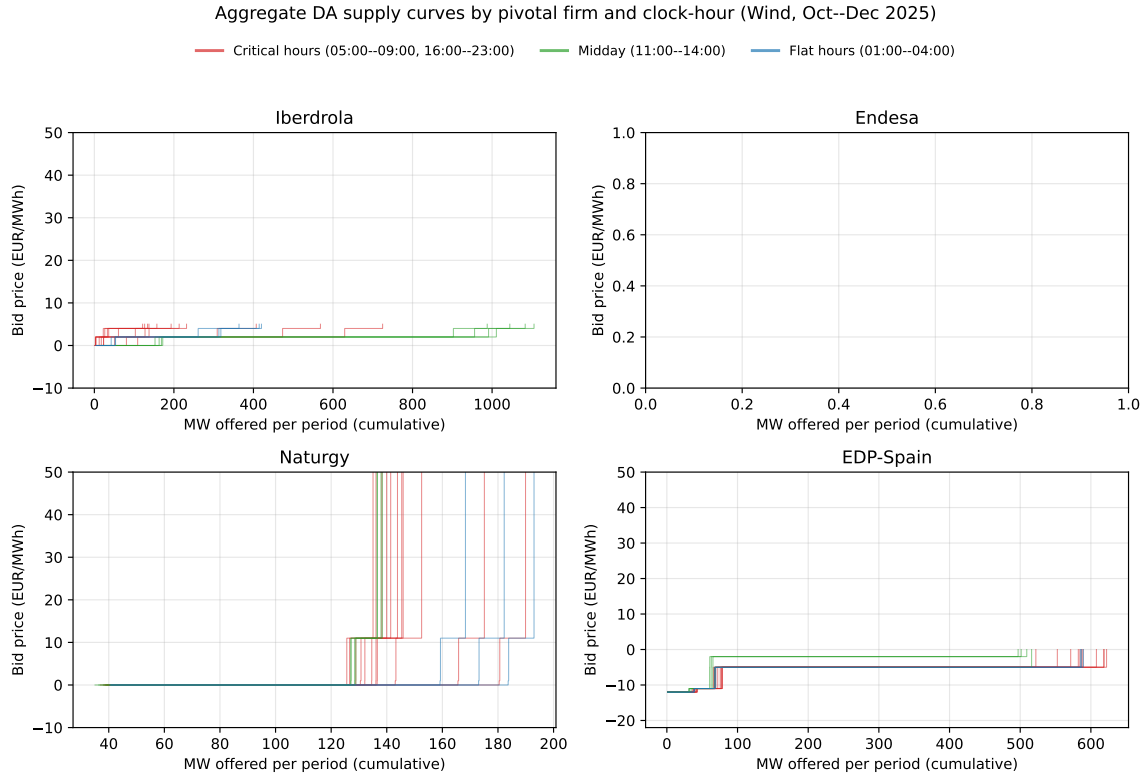


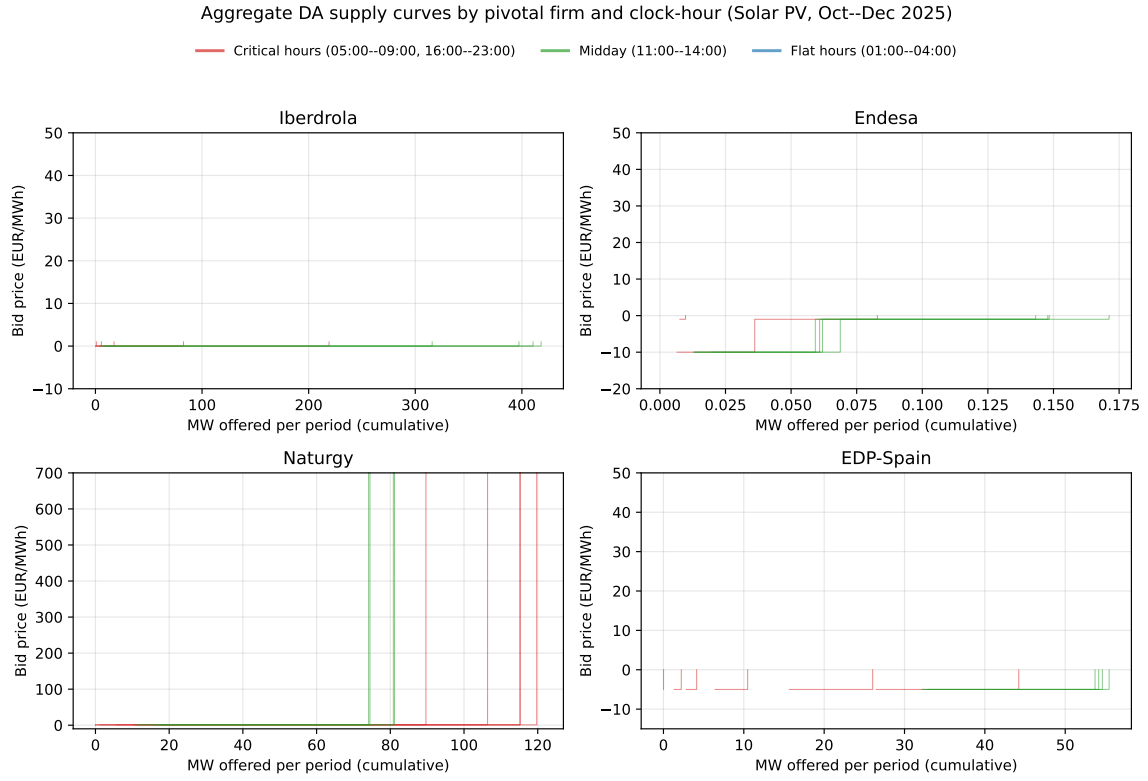
Figure 8: Aggregate DA supply curves, Hydro, October–December 2025. Same construction as Figure 4.

*Reading (mixed).* Iberdrola and Naturgy show clear within-class spread of red curves — hydro is reservoir-managed but still bid-shaped within the day. Endesa's hydro is tiny and concentrated at a few capacity bins. EDP-Spain hydro is essentially a single curve across all hours: no within-day differentiation.



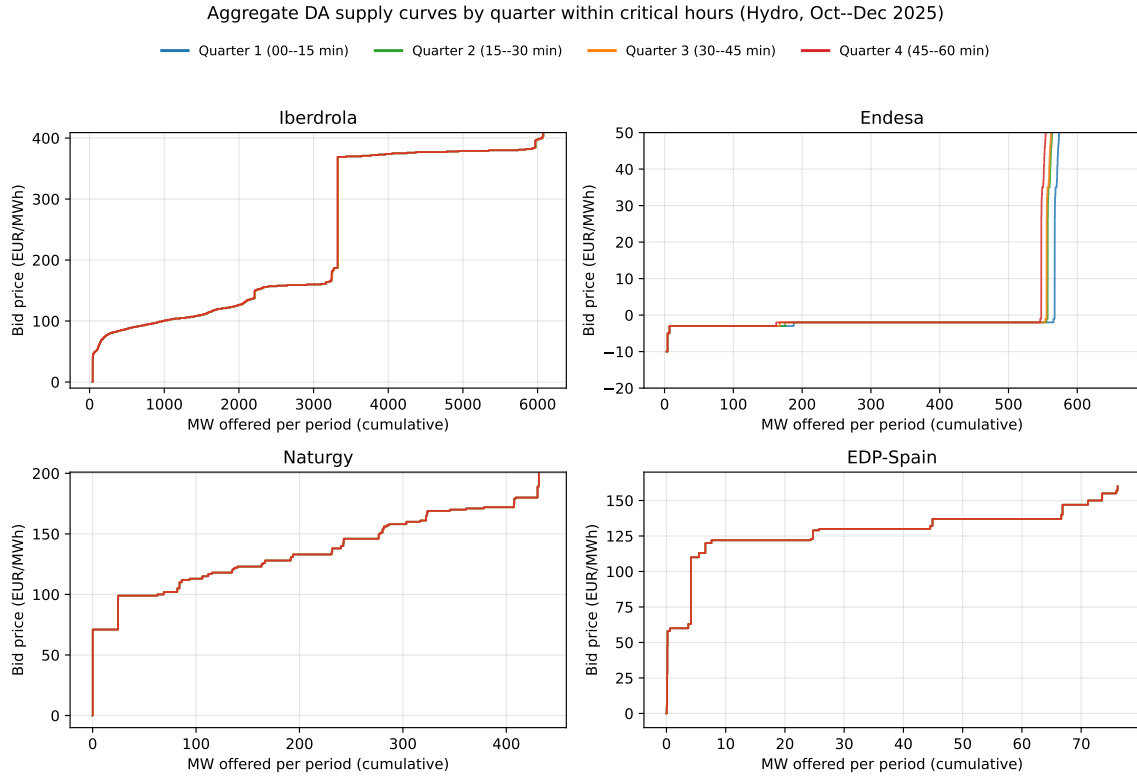
**Figure 9: Aggregate DA supply curves, Wind, October–December 2025.** Same construction as Figure 4.

*Reading (supportive).* Wind units bid at the price floor as expected of a price-taker technology; the curves are essentially horizontal. Cross-hour spread is minimal – consistent with the prediction that intermittent renewables do not exploit within-hour granularity.

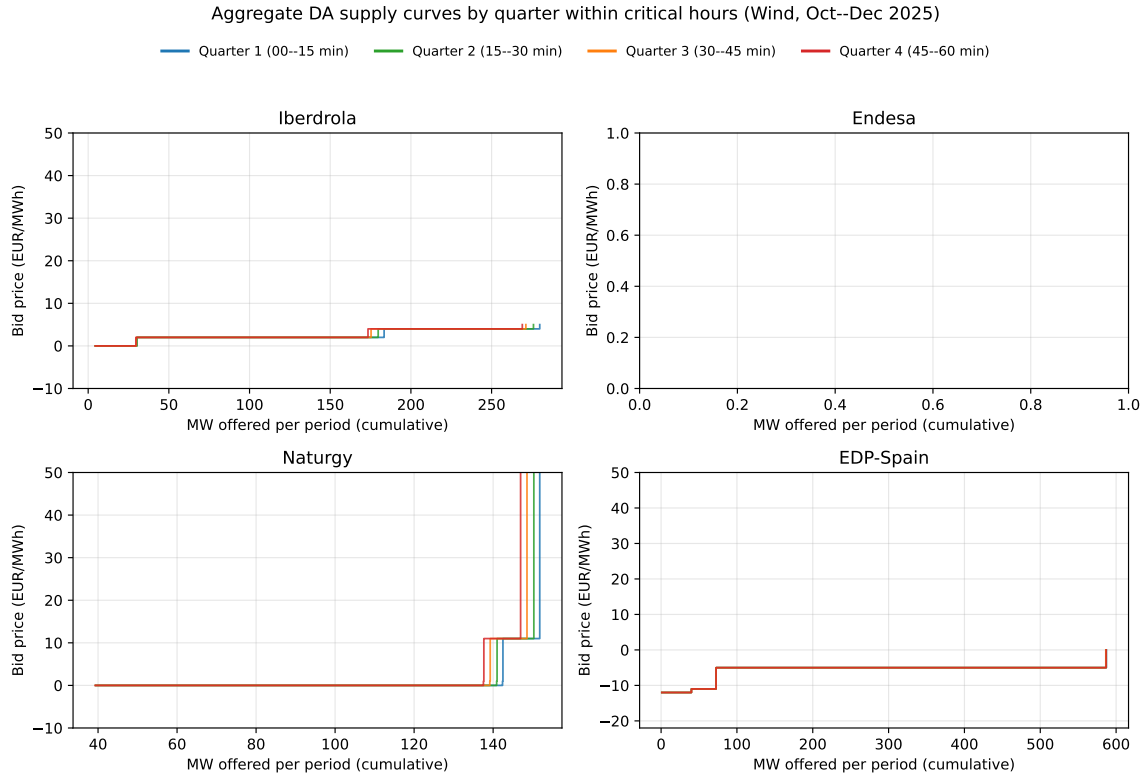


**Figure 10: Aggregate DA supply curves, Solar PV, October–December 2025.** Same construction as Figure 4.

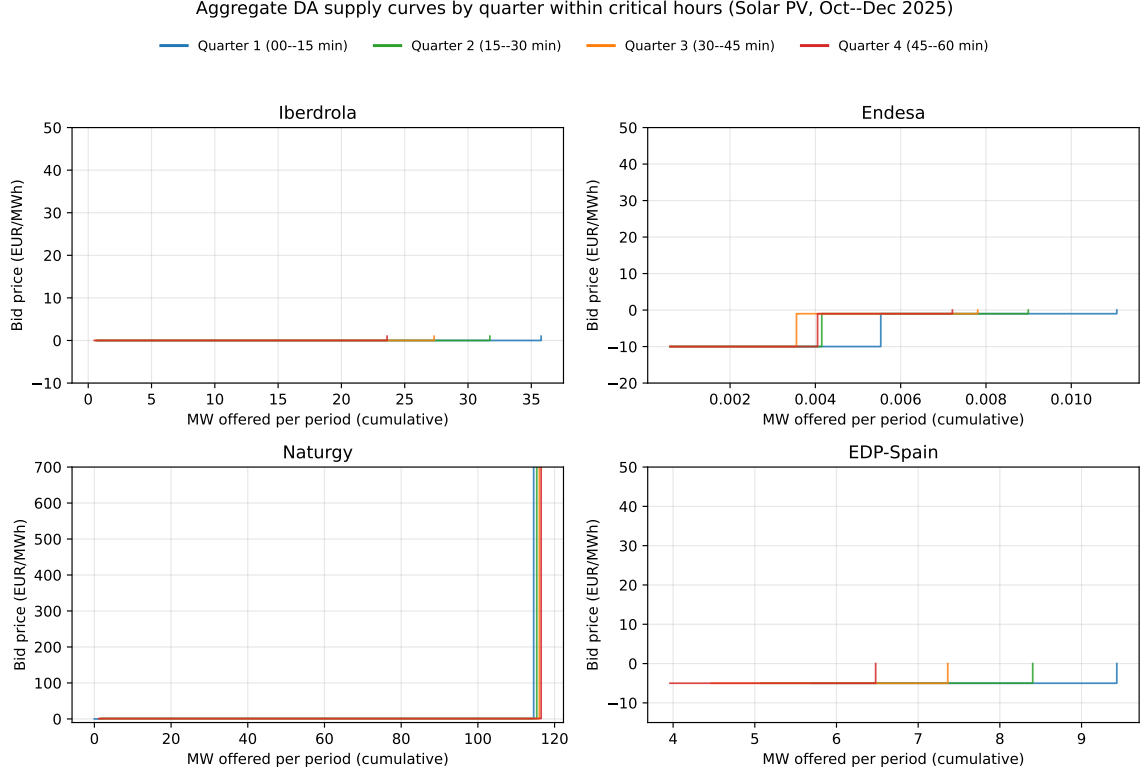
*Reading (supportive).* Same picture as Wind: solar PV bids cluster at the price floor with negligible within-day variation. The Endesa panel is essentially empty – under the primary-owner allocation Endesa retains little solar capacity in our sample window. Nuclear and Coal are omitted: under primary-owner allocation only one pivotal firm holds the assets (Ascó / Vandellòs → Endesa; Soto de Ribera coal → EDP-Spain), so the within-firm comparison degenerates.



**Figure 11: Aggregate DA supply curves by quarter, Hydro.** Same construction as Figure 5, restricted to critical hours.



**Figure 12: Aggregate DA supply curves by quarter, Wind.**



**Figure 13: Aggregate DA supply curves by quarter, Solar PV.**

*Reading (supportive).* Across Hydro, Wind and Solar PV the four quarter-curves are visually indistinguishable for every firm. None of the non-CCGT technologies of pivotal firms exploits the within-hour granularity of the day-ahead market.

### A.11 Time placebos

### A.12 Triple-difference: pivotality as moderator

$$\begin{aligned}
 q_{2,fudh} = & \alpha + \beta_1 \text{crit}_h + \beta_2 \text{post}_d + \beta_3 \text{piv}_f \\
 & + \beta_{12} \text{crit}_h \text{post}_d + \beta_{13} \text{crit}_h \text{piv}_f + \beta_{23} \text{post}_d \text{piv}_f \\
 & + \beta_{123} \text{crit}_h \text{post}_d \text{piv}_f + \gamma_f + \delta_{\text{DOW}(d)} + \boldsymbol{\theta}' X_{dh} + \varepsilon_{fudh}, \quad (4)
 \end{aligned}$$

where  $X_{dh} = (\text{wind}_{dh}, \text{solar}_{dh}, \text{demand}_{dh})$  in GW.

**Table 10: Triple-difference on  $q_2$ .** Equation 4; pivotal and non-pivotal firms pooled. SE clustered by date.

|                                       | (1)<br>DDD<br>baseline | (2)<br>+ X<br>(additive) | (3)<br>+ X×{crit,<br>post, piv} | (4)<br>+ clock-hour<br>FE |
|---------------------------------------|------------------------|--------------------------|---------------------------------|---------------------------|
| $\beta_{123}$ (crit×post×piv)         | 8.605<br>( 1.362)      | 8.623<br>( 1.364)        | 8.612<br>( 1.361)               | 8.619<br>( 1.364)         |
| Implied $\beta_3$ , pivotal firms     | 3.862<br>( 1.146)      | 3.874<br>( 1.147)        | 4.006<br>( 1.154)               | 3.875<br>( 1.148)         |
| Implied $\beta_3$ , non-pivotal firms | -4.743<br>( 0.423)     | -4.748<br>( 0.422)       | -4.605<br>( 0.488)              | -4.743<br>( 0.422)        |
| Observations                          | 293,838                | 293,838                  | 293,838                         | 293,838                   |

*Reading (supportive).*  $\beta_{123} \approx +8.6$  MWh (SE  $\approx 1.4$ ) across all four specifications, with  $p < 0.001$ . The covariates and clock-hour fixed effects move the coefficient by less than 0.02 MWh, so the result does not rely on the day-level controls.

**Table 11: Time-placebo  $\beta_3$  on  $q_2$ .** Equation 2 re-estimated over windows not crossing the October 2025 reform. P1/P2 are within-regime placebos; P3 crosses the June 2024 intraday reform.

|                               | P1: within-2024  |                    |                     | P2: within-2025 pre-reform |                  |                      | P3: shifted back 1 year |                    |                  |
|-------------------------------|------------------|--------------------|---------------------|----------------------------|------------------|----------------------|-------------------------|--------------------|------------------|
|                               | All              | Pivotal            | Non-piv.            | All                        | Pivotal          | Non-piv.             | All                     | Pivotal            | Non-piv.         |
| Critical × Post ( $\beta_3$ ) | 0.942<br>(0.490) | -0.0305<br>(1.235) | 1.910***<br>(0.440) | -0.0637<br>(0.407)         | 1.200<br>(0.894) | -1.133***<br>(0.280) | -0.916<br>(0.536)       | -2.785*<br>(1.341) | 0.199<br>(0.439) |
| Obs.                          | 253,079          | 98,092             | 154,987             | 327,300                    | 146,896          | 180,404              | 250,320                 | 99,636             | 150,684          |
| Clusters                      | 184              | 184                | 184                 | 182                        | 182              | 182                  | 184                     | 184                | 184              |

*Reading (supportive).* P1 (within-2024) and P2 (within-2025 pre-MTU15-DA) give  $\beta_3$  statistically indistinguishable from zero in the pivotal sample. P3, which crosses the June 2024 intraday reform, picks up a negative coefficient consistent with that earlier structural shift, not the October 2025 reform.

## B Data appendix

### B.1 Sequential-market technical reference (PDBC, PDBF, PDVD, PIBCA, PHF)

### B.2 Firms in the analysis sample (detailed)

I THINK WE CAN TAKE THE CAPACITY DATA! BUT MAYBE IT IS ONLY FOR EXPORTS AND IMPORTS!.

Column definitions. *DA auction share*: PDBF offer-type 1, anonymous DA clearing. *DA bilateral share*: PDBF offer-type 4, pre-negotiated firm-to-firm transfers. *IDA cumulative share*: PIBCA accumulated (last session per unit-period). *Net position*: net seller if 2025 DA sales  $> 60\%$  of turnover, net buyer if  $< 40\%$ , mixed otherwise. The net-position label is descriptive; vertical integration does not enter the identification.

**Table 12: Firms in the analysis sample (detailed).** Channel-segregated 2025 shares. Channel denominators are channel-specific (they do not sum across rows).

| Parent firm                                                 | Gen.<br>units | Capacity<br>(GW) | DA auction<br>share (%) | DA bilateral<br>share (%) | IDA cumul.<br>share (%) | Net position |
|-------------------------------------------------------------|---------------|------------------|-------------------------|---------------------------|-------------------------|--------------|
| <i>Dominant operators in Spanish generation (CNMC tier)</i> |               |                  |                         |                           |                         |              |
| Iberdrola                                                   | 70            | 31.03            | 20.01                   | 27.18                     | 20.37                   | Mixed        |
| Endesa                                                      | 38            | 12.46            | 36.88                   | 7.11                      | 12.64                   | Net buyer    |
| Naturgy                                                     | 39            | 12.60            | 3.86                    | 16.18                     | 7.67                    | Net buyer    |
| EDP-Spain                                                   | 21            | 4.73             | 0.77                    | 5.67                      | 2.60                    | Net buyer    |
| EDP-Portugal                                                | 16            | 7.42             | 20.62                   | 12.70                     | 5.17                    | —            |
| <i>Other large operators (non-dominant)</i>                 |               |                  |                         |                           |                         |              |
| Repsol                                                      | 13            | 0.61             | 1.79                    | 1.26                      | 0.62                    | Net buyer    |
| Engie España                                                | 59            | 4.05             | 0.03                    | 2.49                      | 1.07                    | Net seller   |
| TotalEnergies                                               | 15            | 1.19             | 1.19                    | 0.38                      | 0.59                    | Net buyer    |
| Moeve                                                       | 112           | 1.28             | 0.03                    | —                         | 0.66                    | Net buyer    |

*Reading (descriptive).* Iberdrola is roughly balanced across channels. Endesa is auction-heavy. Naturgy is bilateral-heavy. Vertical-integration status is descriptive only and does not enter the identification.

### B.3 File-format specifications and parser conventions

### B.4 Power-vs-energy unit discipline

### B.5 Coverage and missingness

### B.6 Hour-of-day classification metrics

**Table 13: Hour-of-day classification metrics, Spain 2025.** Mean load, solar, net-load (= load – wind – solar), within-hour SD of net-load, and signed within-hour load change  $\Delta\text{load} = \text{load}_{q_4} - \text{load}_{q_1}$ , by clock-hour.

| Clock-hour  | Demand (GW) | Solar (GW) | Wind (GW) | Residual demand (GW) | $\sigma_{\text{within}}$ (MW) | $\Delta$ demand (MW) | Class           |
|-------------|-------------|------------|-----------|----------------------|-------------------------------|----------------------|-----------------|
| 00:00–01:00 | 23.2        | 0.2        | 7.1       | 16.0                 | 405                           | –864                 | <i>dropped</i>  |
| 01:00–02:00 | 22.4        | 0.1        | 6.8       | 15.4                 | 353                           | –481                 | <i>flat</i>     |
| 02:00–03:00 | 21.9        | 0.1        | 6.6       | 15.1                 | 269                           | –193                 | <i>flat</i>     |
| 03:00–04:00 | 21.9        | 0.1        | 6.5       | 15.3                 | 324                           | 194                  | <i>flat</i>     |
| 04:00–05:00 | 22.9        | 0.1        | 6.4       | 16.4                 | 649                           | 1179                 | <i>dropped</i>  |
| 05:00–06:00 | 24.9        | 0.3        | 6.3       | 18.3                 | 766                           | 1795                 | <i>critical</i> |
| 06:00–07:00 | 27.0        | 1.9        | 6.2       | 18.9                 | 1009                          | 1420                 | <i>critical</i> |
| 07:00–08:00 | 28.3        | 5.9        | 5.8       | 16.7                 | 1259                          | 602                  | <i>critical</i> |
| 08:00–09:00 | 28.7        | 10.5       | 5.3       | 12.9                 | 1116                          | 218                  | <i>critical</i> |
| 09:00–10:00 | 28.7        | 13.6       | 5.0       | 10.1                 | 783                           | –103                 | <i>dropped</i>  |
| 10:00–11:00 | 28.6        | 15.0       | 4.8       | 8.7                  | 576                           | –78                  | <i>dropped</i>  |
| 11:00–12:00 | 28.6        | 15.6       | 4.9       | 8.1                  | 471                           | 116                  | <i>dropped</i>  |
| 12:00–13:00 | 28.7        | 15.6       | 5.1       | 8.0                  | 448                           | –60                  | <i>dropped</i>  |
| 13:00–14:00 | 28.3        | 15.2       | 5.3       | 7.9                  | 453                           | –290                 | <i>dropped</i>  |
| 14:00–15:00 | 28.0        | 14.4       | 5.5       | 8.1                  | 545                           | –57                  | <i>dropped</i>  |
| 15:00–16:00 | 28.3        | 13.0       | 5.8       | 9.4                  | 747                           | 344                  | <i>dropped</i>  |
| 16:00–17:00 | 28.9        | 10.1       | 6.3       | 12.5                 | 1241                          | 634                  | <i>critical</i> |
| 17:00–18:00 | 30.0        | 6.4        | 6.8       | 16.8                 | 1362                          | 902                  | <i>critical</i> |
| 18:00–19:00 | 31.2        | 2.7        | 7.3       | 21.2                 | 1185                          | 847                  | <i>critical</i> |
| 19:00–20:00 | 32.0        | 0.7        | 7.4       | 23.9                 | 557                           | 163                  | <i>critical</i> |
| 20:00–21:00 | 31.1        | 0.3        | 7.5       | 23.2                 | 614                           | –1487                | <i>critical</i> |
| 21:00–22:00 | 28.6        | 0.2        | 7.5       | 20.9                 | 801                           | –1926                | <i>critical</i> |
| 22:00–23:00 | 26.4        | 0.2        | 7.4       | 18.8                 | 694                           | –1441                | <i>critical</i> |
| 23:00–00:00 | 24.6        | 0.2        | 7.2       | 17.2                 | 534                           | –1093                | <i>dropped</i>  |

*Reading (supportive).* Critical hours (05:00–09:00, 16:00–23:00) have within-hour SD of net-load several times higher than flat hours (01:00–04:00). The ranking pins the classification directly to the structural variability of residual demand.

## References

- Alberto Abadie. Semiparametric difference-in-differences estimators. *The Review of Economic Studies*, 72(1):1–19, 2005.
- Blaise Allaz and Jean-Luc Vila. Cournot competition, forward markets and efficiency. *Journal of Economic Theory*, 59(1):1–16, 1993.

- James B. Bushnell. A mixed complementarity model of hydro-thermal electricity competition in the western US. *Operations Research*, 51(1):80–93, 2003.
- Clément de Chaisemartin and Xavier D’Haultfoe uille. Two-way fixed effects estimators with heterogeneous treatment effects. *American Economic Review*, 110(9):2964–96, 2020.
- James J. Heckman, Hidehiko Ichimura, and Petra E. Todd. Matching as an econometric evaluation estimator: Evidence from evaluating a job training programme. *The Review of Economic Studies*, 64(4):605–654, 1997.
- Ali Hortaçsu and Steven L. Puller. Understanding strategic bidding in multi-unit auctions: a case study of the texas electricity spot market. *RAND Journal of Economics*, 39(1):86–114, 2008.
- Koichiro Ito and Mar Reguant. Sequential markets, market power, and arbitrage. *American Economic Review*, 106(7):1921–57, 2016.
- Erin T. Mansur. Measuring welfare in restructured electricity markets. *Review of Economics and Statistics*, 90(2):369–386, 2008.
- Pedro H. C. Sant’Anna. Causal inference using difference-in-differences, lecture 5: How covariates can make your did more plausible. Emory University, January 2025, 2025.
- Pedro H. C. Sant’Anna and Jun Zhao. Doubly robust difference-in-differences estimators. *Journal of Econometrics*, 219(1):101–122, 2020.
- Frank A. Wolak. Measuring unilateral market power in wholesale electricity markets: the california market, 1998–2000. *American Economic Review*, 93(2):425–430, 2003.